CHAPTER 23: PALLIATIVE CARE

- Palliative care is an approach that improves the quality of life of patients and their families facing the problem associated with life-threatening illness through the prevention and relief of suffering by means of early identification and impeccable assessment and treatment of pain and other problems, physical, psychosocial and spiritual.

Palliative care:

- Provides relief from pain and other distressing symptoms;
- Affirms life and regards dying as a normal process;
- Intends neither to hasten nor postpone death;
- Integrates the psychological and spiritual aspects of patient care;
- Offers a support system to help patients live as actively as possible until death;
- Offers a support system to help the family cope during the patient's illness and in their own bereavement;
- Uses a team approach to address the needs of patients and their families, including bereavement counseling, if indicated;
- Will enhance quality of life, and may also positively influence the course of illness;
- Is applicable early in the course of illness, in conjunction with other therapies that are intended to prolong life, such as chemotherapy or radiation therapy, and includes those investigations needed to better understand and manage distressing clinical complications.

23.1. Pain management

Section A: Causes of Pain, Assessment and Treatment

The International Association for Study of Pain (IASP) defines pain as an unpleasant sensory and emotional experience associated with actual or potential tissue damage or described in terms of such damage.

The concept of total pain refers to the global nature of pain perception not only as a physical ailment but that it has a psychological, spiritual and social consequences.

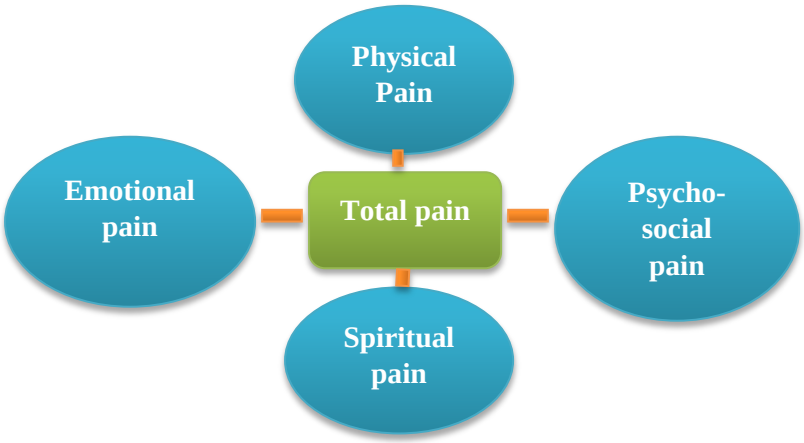


Figure: 23.1. The concept of Total Pain (first described by Cecily Saunders)

Cause and classifications of pain

- **Acute versus chronic pain**
  - **Acute pain** is due to a definable acute disease or injury, duration limited to healing of tissue in days or weeks, accompanied by anxiety and sympathetic over-activity (sweating, tachycardia, tachypnea).
  - **Chronic pain** on the other hand persists months beyond the usual course of an acute disease reasonable time for an injury to heal, or is associated with a chronic pathological process which causes continuous pain or pain which recurs at intervals for months or years. It is accompanied by psychological problems such as depression but no sympathetic response due to adaptation of the sympathetic nervous system.
- **Neuropathic versus Nociceptive Pain**
  - **Neuropathic pain** is caused by damage to the central or peripheral nervous system. Neuropathic pain can be caused by injury, compression or infiltration of a nerve. Often neuropathic pain is described as a burning, tingling or stinging sensation or a shooting electric shock-like sensation – 'pins and needles'.
  - **Nociceptive pain** arises from noxious stimuli, potential or actual injury of somatic or visceral tissues of the body.

Assessment of Pain

Pain should be considered as the **5th vital sign**. A proper assessment of pain is essential for successful treatment. The initial assessment involves the following: detailed history taking including psychosocial assessment, physical assessment, and diagnostic evaluation.

In palliative care there are many different possible causes of pain. It is important to establish the cause of pain. Therapy should be directed at the cause wherever possible. The best approach to differentiate cause and origin is to attempt to characterize pain through the well-known pneumonic **"P Q R S T".**

- **P** refers to Precipitating and palliating (relieving) factors
- **Q** refers to Quality of pain (e.g. burning, stabbing, throbbing, aching, stinging)
- **R** to Radiation of pain
- **S** to Site (document on body diagram) and
- **T** to Timing (duration of pain, recurrence, whether constant or intermittent) and Treatment (the effect of current and previous medications)

In palliative care it is not only sufficient to characterize the pain but also measure the degree of severity as objectively as possible. To assess success of treatment some form of quantitation of pain is necessary.

The simplest and most reliable index of pain is the patient's verbal report. For regular follow-up, it will be useful to grade the pain at every visit on a pain scale. Many pain-scoring systems (pain scales) are available, but the numerical scale of ten is the most recommended in adults while the facial scale is effective in pediatric age.

Scales

- *Categorical or verbal rating scale*: A four or five point scale could grade the pain as none, mild, moderate, severe and excruciating etc. This scale is simple to apply but is not sensitive enough.
- *Numerical Scale*: It has 0 at one end meaning no pain and 10 at the other end meaning worst imaginable pain. (No pain) **0** 1 2 3 4 5 6 7 8 9 **10** (Worst possible pain)
- *Visual Analogue Scale (VAS):* - A100 mm scale with no pain at one end worst imaginable pain at the other is particularly used in clinical trials, as it is more amenable to statistical analysis than numerical scale. It is a simple line on which the patient marks **X** to denote how strong their pain is

No pain ——————|——————— Worst possible pain

- Palms Pain Scale or Five-finger score (0-5)



The hand scale

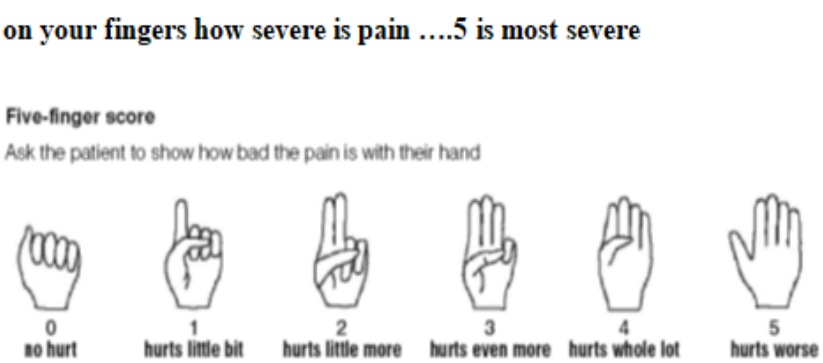


Figure: 23.2. Five-finger score

The hand scale ranges from a clenched hand (which represents 'No hurt') to five extended digits (which represents 'Hurts worst'), with each extended digit indicating increasing levels of pain.

Note: it is important to explain this to the patient as a closed fist could be interpreted as worst possible pain in some cultures.

- For children: *The Faces pain scale*



Figure: 23.3. The Faces pain scale for children

(The Faces pain scale has been revised so that the scale is from zero to ten).

- Use with children who can talk (usually 3 years and older)
- Explain to the child that each face is for a person who feels happy because he has no pain, or a little sad because he has a little pain, or very sad because he has a lot of pain.
- Ask the child to pick one face that best describes his or her current pain intensity.
- Record the number of the pain level that the child reports to make treatment decisions, follow-up, and compare between examinations.
- For use in children less than three years of age or older non-verbal children Face pain scale is used to assess pain.

Principles of chronic pain management

- The management of chronic pain involves the use of oral drug therapy by the clock depending on the duration of action of the drug.
- Pain should be treated whenever possible, consider pharmacologic precautions for all analgesic drugs.
- **'By the mouth':** The oral route is best for the management of chronic pain. Oral medications should only be abandoned if the patient is unable to take or retain them.
- **'By the clock':** Analgesics should be given *'by the clock'*, i.e. at regular intervals. Analgesics are given according to a strict schedule determined by the duration of action, in order to prevent recurrence of pain. A patient on a regular schedule of analgesia will also need to have available a 'breakthrough' dose for any episodes of breakthrough pain.
- **'By the ladder':** the two/three step analgesic ladder (two step for pediatric, three step for adults) which is meant for mild, moderate and severe pain is to be followed. Unless the patient is in severe pain, begin by prescribing a non-opioid drug and adjust the dose, if necessary, to the maximum recommended dose.
- **'For the individual':** the right dose of an analgesic is the dose that relieves the pain.
- **'Attention to detail':** it is essential to monitor the patient's response to the treatment to ensure that the patient obtains maximum benefit with as few adverse effects as possible
- **'Combination therapy'** using two or more analgesia of different mechanism of action is evidence-based practice.

WHO Analgesic Ladder

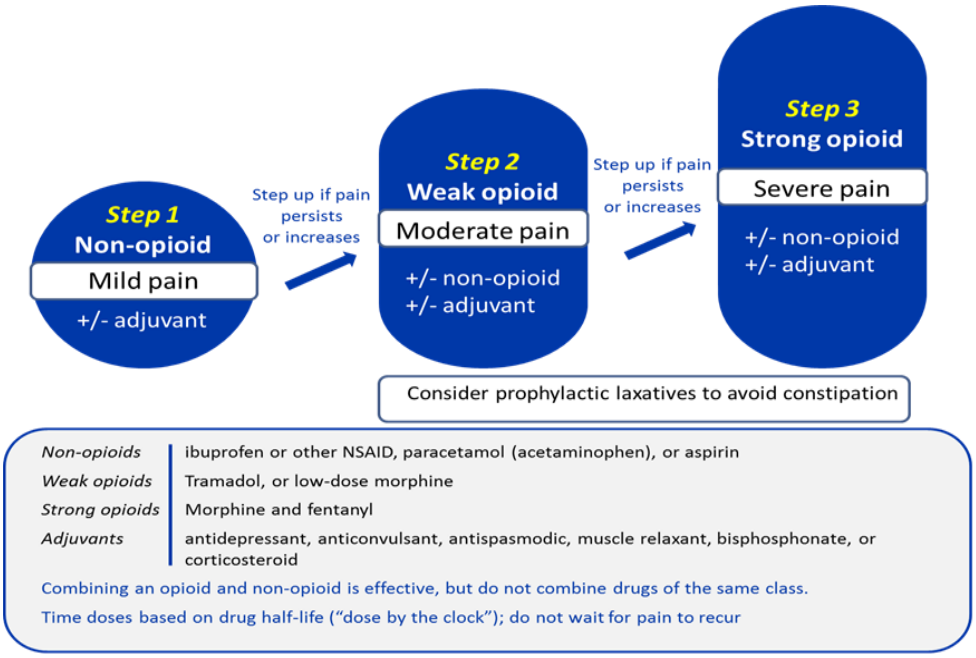


Figure: 23.4. The World Health Organization 3-Step Analgesic Ladder for Adults

Step I: Non opioid analgesia

NSAIDs: They have a key role in the management of pain associated with inflammation as in soft tissue infiltration and bone metastases. NSAIDs differ in their effect on platelet function. Aspirin causes irreversible platelet inhibition whereas Ibuprofen, etc. causes reversible platelet inhibition. Diclofenac etc. do not have any commendable effect on platelet function.

The dosage schedule for commonly used NSAIDs is:

- **Diclofenac**: 50mg PO TID,
- **Ibuprofen:** 400-600mg TID,
- **Paracetamol**: 1 gm Po QID

Step II: Weak Opioids

Tramadol: 50-100mg q8h-q6h. Maximum dose is 400mg.

Weak opioids are not recommended for pediatrics age group (see 2 step analgesic ladder)

Low-dose morphine can also be used at Step 2.

Step III- Strong Opioids

The dose must be titrated for each individual patient starting from the lowest possible dose in elderly and cachectic patients of 2.5mg/4hourly to normal adults of 5mg/4hourly. There is no standard dose for morphine for the treatment of chronic or cancer related pain and hence no ceiling effect, meaning dose is escalated 72 hours and progressively until pain is controlled and side effects such as nausea and drowsiness are tolerated. Always prescribe a laxative along with an opioid regularly as tolerance is never developed for this particular, and, in selected patients an anti-emetic on as required basis is prescribed if the risk of nausea and vomiting is very disturbing.

Normal release morphine (NR) is always started on four hourly basis and begins to work after about 20 minutes and analgesia lasts four hours. In an opioid naïve patient, start with 2.5-5 mg morphine. Experienced Patients, who are receiving an opioid, may require higher doses (based on equianalgesic doses). It should be given four hourly. A double dose is given at bedtime and the midnight dose is skipped. A rescue dose is advised for breakthrough pain. This should be the same as the four hourly dose of morphine. If these breakthrough episodes persist and become more than 4 /day, dose adjustment as above is required. The dose is increased by 30-50% every 3 days and it can also be reduced progressively in the same manner.

Adverse effects: Even with the therapeutic dose of opioids the following adverse effects are common.

- Constipation occurs in about 95% of patients and it may last as long as the drug is continued. Constipation should be prevented rather than treated and co- administration of a laxative is a must. Bisacodyl 5mg at night, increasing to 15mg if needed) unless the patient has diarrhea.
- Nausea and vomiting occurs in 1/3rd of patients. It is usually seen in the first few days of therapy and is usually self-limiting. Treat with Metoclopramide 10mg tid or Haloperidol 1.5mg once a day.
- Itching is also seen in less than 7% of the patients sometimes respond to 2-3 days of antihistamine therapy, use Chlorpheniramine.
- Other side effects include dry mouth, urinary hesitancy, and sleepiness.

Signs of toxicity: These appear when the administered dose of morphine is more than what is required for pain relief, or when the pain is not morphine-responsive, yet dose is escalated progressively. The signs are toxicity are delirium, myoclonus and drowsiness. Drowsiness occurs in up to 1/3rd of patients on initiation of treatment or following a significant dose increase. If it persists dose reduction is needed. Dose reduction is recommended in renal impairment and old age and debility. These toxicities can be effectively managed with specific drugs including naloxone, an opioid antagonist.

As long as the dose escalation for morphine is made in a stepwise manner, there is less likelihood of an excessive dose being given toxicities are unlikely to occur. But in situations such as renal failure there may be accumulations resulting in toxicities. There are reports where compulsive drug seeking behaviors exhibited by few patients who have a past history of psychiatric illness. With the same token respiratory depression and addiction are NOT also problem with oral morphine in patient with a clear indication. On the other hand, physical dependence is a normal physiological response to opioid therapy, which causes withdrawal symptoms, if the drug is abruptly stopped or an opioid antagonist is administered. Withdrawal symptoms can be effectively managed by gradual and supervised reduction of the opioid therapy.

Naloxone – reverses all opioid side effects, so both respiratory depression and pain relief are reversed. Too much naloxone given too quickly and reversing analgesia may result in restlessness hypertension and arrhythmias and has been known to precipitate cardiac arrest in a sensitive patient.

- Indications for Naloxone
  - RR < 8/minute.
  - RR <12/minute, difficult to rouse, cyanosis
  - RR < 12/minute, difficult to rouse, SaO2 <90%

Dose: dilute naloxone 400 micrograms to 10ml with 0.9% saline. Give 0.5ml (20 micrograms) IV every 2min until respiratory status is satisfactory. Further boluses may be necessary because naloxone is shorter-acting than morphine.

Note: There can be concerns when prescribing morphine that addiction can occur. When used correctly, patients do not become dependent, tolerance is uncommon and respiratory depression does not usually occur. The risk of addiction is commonly overestimated by patients and family, so it is important for healthcare workers to alleviate their fears. When discontinuing morphine, avoid symptoms of withdrawal by titrating the opioid dose down slowly.

WHO Analgesic Ladder for Children

The two-step approach is an effective strategy for the pharmacological management of persisting pain in children rather than the three-step analgesic ladder used for adults. The two-step approach considers the use of low doses of strong opioid analgesics for the treatment of moderate pain as the benefits of using an effective strong opioid analgesic outweigh the benefits of intermediate potency opioids in the pediatric population and although recognized, the risks associated with strong opioids are acceptable when compared with the uncertainty associated with the response to codeine and tramadol in children.

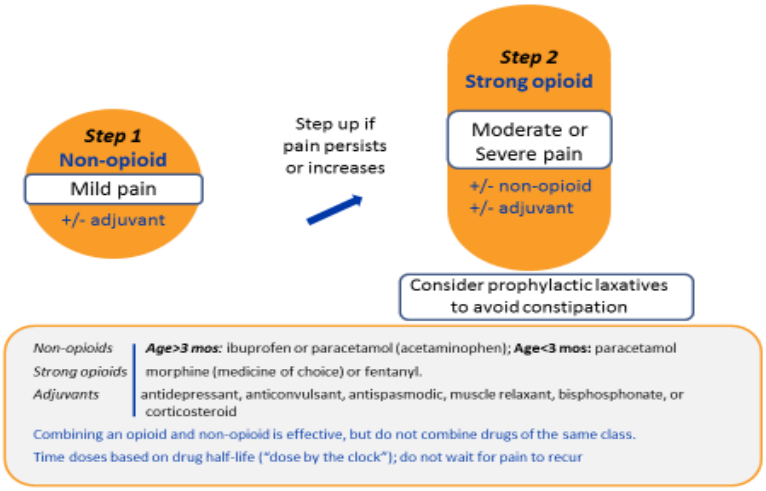


Figure: 23.5. The World Health Organization 2-Step Analgesic Ladder for Children

Variation from the WHO pain management protocol

- By mouth administration may not be feasible in dysphagia (sub-lingual or per-rectal or parenteral administration).

- By the clock administration may not be possible as well in cases with renal failure and dyspnea where the dose and frequency should be reduced or spaced. In renal failure, morphine interval may need to be longer or the dose reduced. If no urine output, then morphine should be stopped and given PRN.
- For breakthrough pain – the drug has to be given in between the specified intervals.
- In neonates – half life and clearance are variable.
- A double dose is used at bed time to avoid waking the patient for medication
- By the ladder administration may be difficult in the following instances:
    - As in morphine trial for quick pain relief and for assessing the dose requirement.
    - Sticking to step 2 weak opioids when morphine is not available.
    - Morphine may be used in smaller doses if step 2 are not available.
    - Non opioid may be omitted for poor side effect profile.
    - Adjuvants may be omitted to minimize the number of drugs.
    - Go beyond step 3 in pain that is not responsive

Adjuvants for pain

The adjuvants are not analgesics in the true pharmacological sense, but may contribute significantly to pain relief whether used alone or in combination with the analgesics on the WHO 2/3-step ladder and they may have an analgesic sparing effect. Adjuvant analgesics are of particular use in pain that is partially opioid-sensitive pain. Pain that is less sensitive to opioids includes neuropathic pain, bone pain, pain associated with inflammation and sepsis. Pain associated with smooth or skeletal muscle spasm will not respond to an opioid and will need an adjuvant analgesic. Pain related to anxiety will also benefit from adjuvant analgesics.

- ***Anti-Depressants and Anti-Convulsants:*** Helpful for neuropathic pain, which may present as burning, pricking, allodynia, paresthesia or sharp, shooting pain.
  Amitriptyline: 25mg PO at night. Increase dose as needed.
  OR
  Carbamazepine: 200 mg PO at night. Increase dose as needed
  OR
  Gabapentin: 300 mg PO at night. Increase dose as needed.
- ***Corticosteroids*:** are used as adjuvant treatment if neuropathic pain is suspected to be due to nerve compression e.g. by tumor or inflammation
  Dexamethasone: 4 mg daily
  OR
  Prednisone: 40-60mg daily
- ***Muscle relaxants/Anxiolytics*:** are used as an adjuvant for skeletal muscle spasm and anxiety-related pain.
  Diazepam: 5mg orally, 2-3 times per day.
- ***Antispasmodics:*** are helpful in relieving visceral distension pain and colic.
  Hyoscine Butylbromide: 10mg three times /day PO or IM; can be increased to 40mg three times/day

Non-Pharmacological management of pain

Palliative care includes many non-pharmacological ways to manage 'total' pain. These therapies address the physical, psychological, social and spiritual dimensions of pain.
The following are a few holistic non pharmacological ways of treating pain.

- Anything that enhances quality of life can, in turn, relieve pain.
- Listening and empathy.
- Counseling – provides emotional support and practical suggestions.
- Companionship and accompaniment – can help ease pain and increase comfort.
- Activities such as favorite music, games, gardening, memory book – provides meaning and distraction.
- Spiritual/pastoral support and prayer – provides comfort, meaning and hope.
- Positioning – enhances comfort and relieves pressure areas.
- Bathing, grooming, and other care measures – enhances comfort and self-respect.
- Exercise – improves mobility, circulation and skin integrity.
- Massage, therapeutic touch.
- Traditional therapies that are beneficial, healing and comforting.
- Heat/cool applied locally – can reduce swelling and help relaxation.
- Treatments such as radiotherapy can reduce inflammation, pain and tumor size

Physical care II - Symptom management

Symptoms of various illnesses have for a long time been considered 'sign posts', and merely indicators to a disease and not important by themselves. This common practice is illustrated in acute illnesses whereby symptoms vanish with addressing of the underlying illness, for instance, fever and headache in malaria. However, in chronic non-curable diseases, symptoms may persist long after the diagnosis of disease, as a result of chronic complications, and, are not always useful indicators of underlying conditions and only contribute to suffering and poor quality of life to those affected. Hence it is strongly advised to control them. The general approach to symptom control in palliative care includes assessment and investigation for the undiagnosed disease and severity of the symptom, treatment of reversible causes, initiation of disease/symptom-specific medicines and non-drug measures as well as involvement of the patient and family in the management plan.

The common symptoms in palliative care are:

- Fever
- Respiratory Symptoms
- Gastrointestinal Symptoms
- Urinary Symptoms
- Skin and Wound Care

23.2. Fever

Fever is elevation of core temperature above 37.4 degree Celsius as measured at the axilla.

Table: 23.1. Common cause of fever and its management

Common Causes	Management
• Infections • Inflammatory (autoimmune) • Metabolic • Malignancy • Drugs & toxins	• If new fever, consider cause, work up and treat properly • If bacterial infection treat with appropriate antibiotics • If malaria treat with antimalarial • If neoplastic in origin, treat with appropriate chemotherapy • If transfusion related, discontinue and treat and work up • **For adults**: Give paracetamol 1 gm every 6 hours (no more than 8 tablets of paracetamol in 24 hours). • **For children**: Provide paracetamol 10-15mg/kg oral per dose, maximum 6 dose in 24 hours. • **Neonate** 5-10mg/kg per dose max of 4 doses in 24 hours, avoid aspirin. • Alternative to reduce fever are NSAIDs (Ibuprofen, avoid if less than 3 months). • Make sure patient stays hydrated. **Home Care Advice** • The sick person will lose a lot of water through sweating; therefore encourage him or her to frequently • Drink water, diluted tea, fruit juices and make sure patient stays hydrated. • To cool the body temperature, wipe the body with lukewarm water (cloth soaked or give a bath). • Encourage febrile patient to wear only light clothes. • Paracetamol can be used as above • *Advice: Seek help if fever does not improve or recurs after treatment. Also if fever is accompanied by cough, diarrhea, severe pain, confusion, night sweats, rigors, stiff neck or unconsciousness.*

23.3. Respiratory Symptoms

Dyspnea or breathlessness is a subjective sensation that patients describe as chest tightness, breathlessness, air hunger, inability to take a deep breath, a feeling of suffocation or smothering, or an inability to get enough air. The symptoms usually worsen with exertion and often limit the patient's activity.

Table: 23.2. Approach to Reversible causes of dyspnea /shortness of breath

Possible causes of Dyspnea	Classical signs and symptoms	Treatment

Bronchospasm/asthma	Shortness of Breath Non-productive cough Wheezing Decreased air entry	Bronchodilators: Salbutamol Corticosteroids Oxygen therapy
COPD (emphysema/ chronic Bronchitis)	Shortness of Breath Productive Cough Wheezing Decreased air entry	Bronchodilators: Salbutamol Corticosteroids Oxygen therapy Pulmonary rehabilitation Morphine
Congestive Heart Failure	Orthopnea, Proximal Nocturnal Dyspnea, Fatigue, Chest pain, Edema, Inspiratory rales, Elevated Jugular Venous Pressure	Diuretics Cardiac medications *If indicated*, oxygen
Cardiogenic Pulmonary Edema	Severe SOB, Orthopnea, PND, fatigue , Inspiratory rales, Edema, Elevated JVP	Oxygen Diuresis Morphine Treatment of Underlying cause
Bacterial Pneumonia	Cough Fever, chills pleuritic chest pain Localized rales or signs of consolidation	Antibiotics Cautious hydration *If indicated*, oxygen
Tuberculosis	Fever, cough, sweating, chest pain, Weight loss, loss of appetite.	Anti TB, *if indicated* *Corticosteroid* and oxygen
Pleural Effusion	SOB, chest pain Dullness and decreased air entry with radiographic or ultrasound confirmation	Thoracentesis If indicated chest tube insertion Pleurodesis if indicated Management of underlying disease
Upper Airway Compression	Severe shortness of breath Stridor – especially inspiratory Risk factors, such as neck or mediastinal mass	Corticosteroids Small dose Benzodiazepines, Opioids, or Barbiturates for anxiety or sedation *If indicated, consider/refer:* External radiation Tracheostomy/Airway stent
Copious airway secretions/ Bronchiectasis	Coarse crepitation, Ineffective or absent cough	Anticholinergic drugs Antihistamines If indicated antibiotics Rehydration Airway Suctioning
Pulmonary Embolism	Sudden onset of shortness of breath	Anticoagulation: Heparin *with*

	Chest pain Risk factors for venous thromboembolism e.g. like bedridden patients	Warfarin Oxygen Morphine
Lung Interstitial Pneumonia	Shortness of breathing, hepatosplenomegaly, parotid enlargement, cyanosis and clubbing	Oxygen If indicated Corticosteroid Bronchodilators

Note:

- Give small dose oral morphine—this can reduce dyspnea in end-of-life care. Monitor closely but do not let fears of respiratory depression prevent trying this drug.
    - For a patient not on morphine for pain give 2.5 mg.
    - For a patient already on morphine increase the dose by 25%. If this does not work, increase by another 25% till the patient condition improves (read reference on dose limit for dyspnea).

Dyspnea Home Care

In addition to the treatment given by health worker:

- Help the sick person sit in the best position
- Use extra pillows or some back support
- Open windows to allow in fresh air
- Fan with a newspaper or clean cloth
- Give patient water frequently (it loosens sputum)
- Avoid crowding, cooking and smoking in the room of the patient

Management of Pulmonary Secretions

Pulmonary secretions associated with pulmonary infections or chronic bronchitis can produce troubling symptoms for patients, particularly as their increasing weakness and fatigue make coughing exhausting and ineffective. For patients who are still able to cough effectively, interventions should be directed at helping to reduce the exertion required to bring up secretions or reducing excess secretion.

- Use an anti-secretory drug to reduce production of respiratory secretions: Hyoscine butylbromide 20 mg stat and 20–40 mg PO TID.
- Humidified oxygen. Inhaled oxygen is a helpful comfort measure to reduce symptoms of upper airway drying when oxygen is being administered.
- Postural drainage: Chest physiotherapy appropriate to the patient's condition is valuable in managing respiratory secretions. Teach caregivers this technique and encourage them to make a special effort to avoid flat or supine positions that allow pooling of secretions in the pharynx or larynx, and to reposition the patient frequently.
- Hydration: Dehydration can increase sputum viscosity and exacerbate difficulties with expectoration. Hydration—orally or intravenously is useful treatment to this problem.

Cough

Table: 23.3: Common cause of cough and its management

Common Causes	Management
Respiratory infections • Bronchospasm • Bronchial obstruction • Cardiac causes • Drug related like ACE inhibitors • Esophageal reflux • Foreign body aspiration	**Treatment of underlying cause** • Treat infections using appropriate anti-infective agents, evaluate for tuberculosis. • Cough from bronchospasm - often responds to bronchodilators including salbutamol with either inhaled or systemic corticosteroids (if indicated) • In patients who are moving little air with each breath, systemic corticosteroids and frequent nebulization of bronchodilators may help. If symptoms improve and tidal volumes increase, hand-held metered dose inhalers may be effective • Steroids, radiation, Surgery chemotherapy • Diuretics, salt and fluid restriction • Avoid ACE inhibitors • H2 receptor antagonists or proton pump inhibitors may be appropriate (for esophageal reflux). **Symptom management** • Use Dextromethorphan syrup (when indicated) • Give Oral morphine (2.5-5 mg) (when indicated) **Home care (Soothing remedies)** • Spice drinks help to relieve some of the many unpleasant symptoms experienced by patients. • Example: • Cinnamon, ginger, and honey are used to soothe the throat and relieve coughing. • Cinnamon: Add one-quarter teaspoon of cinnamon powder to a cup of clean boiled water (about 150–200 mL). • Add sugar or honey to taste. The drink is ready for use. • Ginger: Add one teaspoon of crushed ginger roots or powder to a cup of clean boiling water. Cover and leave for 5–10 minutes. Add sugar or honey to taste and the drink is ready. Drink as desired. • Honey, ginger, and cinnamon: Add one teaspoon of ginger powder or cinnamon powder to 150 mL honey and stir. • Take 5–10 mL of the mixture 4-hourly for 5 days.

23.4. Gastrointestinal Symptoms

Hiccups

Persistent hiccups are not unusual in terminally ill patients and can be a distracting and distressing symptom. The interruption of normal activity in patients with intractable hiccups can cause depression, sleep deprivation, decreased oral intake, and weight loss.

Table: 23.4: Common cause of Hiccups and its management

Common Causes	Management
• phrenic nerve or diaphragmatic irritation by tumor • Gastric distension • Gastroesophageal reflux, • Severe esophageal candidiasis. • Drugs (benzodiazepines, corticosteroids, barbiturates) • Psychogenic • intracranial mass/lesion	**Treat the underline cause** **Symptomatic management (pediatric dosage)** • Metoclopramide (10 mg tablet, 1-2 tablets three or four times daily 7-10) /dose • For children-0.1-0.2mg/kg/dose 4 times daily • Chlorpromazine 25mg PO tid up to 7-10 days (for children 0.5-1mg/kg) • Haloperidol (5 mg tablet: 1/4 to ½ tablet once to three times daily). • If patient has brain tumor, try antiepileptic medication (like phenytoin). • Drinking cold water, cold water gargling, Valsalva maneuver **Home care** Stimulate the throat: • Quickly eat 2 heaped teaspoons sugar, or • Drink cold water or eat crushed ice, or • Rub with a clean cloth inside the top of the mouth (feel toward the back, where the where the top of the mouth is soft). Interrupt the normal breathing: • Hold breath or breathe into paper bag, stop when you feel uncomfortable. • Pull knees to chest and lean forward (compress the chest).

Nausea and Vomiting

Conduct a detailed history of the patient's illness including duration of symptoms, precipitating events, what has been tried to alleviate symptoms, and what has been most useful in management of these symptoms.

- Evaluate current medications noting those with known GI side effects.
- Assess for possible infectious conditions
- In women in child bearing age, consider pregnancy
- Increased intracranial pressure can present with nausea and vomiting
- Perform a thorough physical exam to localize the symptoms
- Look for surgical causes of nausea and vomiting
- Presence of fever and signs of dehydration (e.g., hypotension) indicate a more serious process.

Table: 23.5: Management of nausea and vomiting

I. Treatment for reversible causes	
Nausea and vomiting secondary to Infection or Drug side effects	- Treat infections with appropriate antimicrobials and Address drug side effects
II. Symptomatic management of nausea and vomiting	
1. Antiemetic	
Dopamine Antagonists	Most commonly used as first line. - The prototype agents for this class are 1. Metoclopramide: 10 mg every 8 hours For children use 0.1-0.2mg/kg/dose 4 times daily 2. Haloperidol & Chlorpromazine which exerts its effects centrally (1-2 mg once daily and 25-50mg every 6-12 hours respectively
Histamine Antagonists	- Promethazine 25mg 1-4 times daily (For children 0.5mg/kg/dose)
Corticosteroids (for intractable vomiting)	- Add Dexamethasone 10mg/m^2 /dose once daily.
Serotonin antagonist (for intractable vomiting)	- Ondansetron 8mg tid
2. Rest Gut	IV fluids -Sometimes it is best to 'rest the gut' by administering intravenous fluids that may contain replacement electrolytes and basal metabolic energy (e.g.40% dextrose) as needed Keeping NPO
3.Reduction of Increased Intracranial Pressure (ICP)	ICP is often an emergency. Treat the Cause - Nursing care-head elevation 30^0 above

	horizontal - Give dexamethasone 4-8mg q8 hours to decrease the pressure as indicated, give Mannitol IV 1-1.5g per kg. - For Cryptococci do lumbar puncture as required
III. Home care	
	• Give clear liquids initially and boil water if not filtered; advance to full liquids as tolerated. • Give small, frequent meals. • Encourage breast feeding in infants and young children • Let the patient select the type of food he prefers Avoid fried and fatty foods. • Keep patients away from the place where food is being cooked (to avoid smell and sight of food). • Encourage patient to be in a sitting position after eating for at least 30minutes. • Seek help from trained health worker for • Vomiting more than once a day, or dry tongue, or passing little passing little urine or abdominal pain.

Dry Mouth (Xerostomia)

Dry mouth (Xerostomia) is a condition whereby salivary flow is reduced due to reduced productionor free flow. This aggravates a dry mouth resulting soreness and difficulty of speech and mastication.

Table: 23.6: Common causes of Dry Mouth (Xerostomia) and its management

Common Causes	Management
• Dehydration, • Mouth breathing • Reduced mastication • HIV-related (OI) infections • anxiety, and depression • Oxygen therapy • Salivary gland diseases • Drugs(antihistamines, anticonvulsants,	**Treat the underline cause** • Treating reversible causes • Treat infections and review drug regimens • Idiopathic or aphthous ulcers can be treated either with a course of Corticosteroids. **Symptomatic management at Home** • Check the mouth, cheeks, palate, gums, tongue, and teeth often to identify and manage any problems early. • Seek help from health worker if dry mouth persists

antidepressants, and anticholinergic) • Radiotherapy which reduce salivary flow	• Take regular sips of water to keep up fluid intake • Keep lips clean, soft, lubricated, and intact as far possible. • Apply petroleum jelly or moisturizing lotion to lips • Brush teeth with a chew stick or a small, soft • Toothbrush (a baby's toothbrush is ideal) after each meal and at night. If available, use fluoride toothpaste. • If brushing is not possible due to pain or bleeding, use soft sponges, cotton buds, or a gloved • finger wrapped with gauze or a soft cloth • Use a mouthwash after each meal and at night (in addition to brushing, not as a substitute). • Avoid a mouth wash that contains alcohol as it dries out the mouth. • Rinse with 15 mL for 60 seconds using alcohol free mouth wash solutions. Suggestions for mouthwash: Saline: 1 teaspoon salt in 500 mL boiled, cooled water Vinegar or lemon juice: 1 teaspoon in one liter of boiled, cooled water. Antimicrobial mouthwashes: 0.2% chlorhexidine gluconate mouthwash Sodium bicarbonate mouthwash if available: 1 teaspoon in 500 mL boiled, cooled water

Nutrition and Weight loss

General Principles

Nutrition support has been shown to benefit palliative care patients by reducing physical deterioration, improving quality of life, and preventing the emotional effect of "starving the patient to death."

- Palliative care patients of all age groups should be educated and encouraged to consume the following food groups (carbohydrates, proteins, vitamins, minerals, fats & oils, dietary fiber and water).
- The successful management of these medicine-food interactions requires understanding clients' individual food access as well as eating habits. Locally available foods are recommended.
- Management of patients shall include assessment and counseling on feeding with regard to the nutritional needs specific to the stage of the illness.
- Patient and carers shall be counseled on appropriate feeding according to the stage of the illness
- Weight loss is an extremely common symptom in chronic illness.

Table: 23.7: Possible Causes of Weight Loss and Malnutrition in chronic illnesses

Mechanisms	Possible causes
Decreased intake of nutrients	• Insufficient resources to procure food (Poverty) • Dysphagia or odynophagia • Anorexia, nausea, or vomiting • Emotional factors (depression, loneliness or grief) • Dementia
Excessive nutrient loss	• Chronic illness • Malabsorption or drug-related diarrhea

Table: 23.8. Management of Symptoms Causing Weight Loss

Symptom/setting	Dietary Advice
Loss of appetite	• Drink high-energy drinks (e.g. milk or yoghurt) • Eat small, frequent meals. • Encourage exercise if possible.
Nausea and vomiting	• Eat small, frequent meals. • Restrict intake of fluids after meals. • Eat cold foods or food at room temperature. • Avoid excessively fatty meals. • Avoid lying down after eating. • See relevant section
Sore mouth/throat	• Eat soft moist foods, (e.g., mashed potato, minced meat). • Use margarine, butter to moisten cooked food (if diarrhea not present). • Avoid sticky foods (e.g., peanut butter). • Avoid dry, rough foods, (e.g. raw vegetables). • Avoid citrus fruits (e.g. lemon, orange, pineapple) and spicy foods. • Eat foods either cold or at room temperature. • See relevant section
Diarrhea	• Eat small, frequent meals. • Drink plenty of isotonic fluids. • Decrease/avoid milk and dairy products. • Fermented dairy products may be tolerated • Decrease high-fat foods • Include foods high in soluble fiber (e.g. bananas, oats). • Avoid caffeine, (e.g. coffee). • Avoid soft drinks (e.g. Mirinda, Coca Cola)

	• Use anti-diarrheal agent in adults if there is no contraindication • See relevant section
Home Care	• Encourage the sick person to eat, but do not use force as the body may not be able to accept it and he or she may vomit. • Offer smaller meals frequently of what the sick person likes. • Accept that intake will reduce as patient gets sicker and during End-of-life care. • Seek help from trained health worker if you notice rapid weight loss or if the sick person consistently refuses to eat any food or is not able to swallow.

- If all other means fail, try Prednisone 5-15 mg daily in the morning to stimulate appetite in end of life care, stop if no effect after 2 weeks.

Dysphagia and Odynophagia

Difficulty of swallowing food as a result of pain or obstruction due to various causes affects the nutritional status and general health of certain chronic diseases.

Table: 23.9: Common cause of Dysphagia and Odynophagia and its management

Common Causes	Management
• Oropharyngeal problems o Candidiasis o Mucositis o Dry mouth o Neurogenic (Stroke, brain tumors or peripheral neuropathies) o Pharyngeal abscess, tumor, tonsillitis • Esophageal problems o Inflammatory/infection (Reflux esophagitis, candida Herpes, CMV, aphthous ulcer, caustic ingestion, radiation mucositis). o Neoplastic (esophageal ca) o Motility disorder(e.g. achalasia)	• Treating reversible causes • Complications like dehydration, electrolyte • disturbance, malnutrition • Symptomatic treatment • Prescribe analgesia • Steroids to reduce inflammation or edema. • If no effect by the third day, stop treatment. • Nasogastric tube,not recommended for long term use due to patient discomfort • Endo-esophageal tube • Gastrostomy tube –contraindicated in Advanced disease Home care • Soft diet to decrease discomfort such as yoghurt or • Depending on what the patient feels is helpful. • Avoid extremely hot or cold or spicy foods. • Upright position to facilitate swallowing

○ Anatomical (hernia, diverticulum stricture) Psychogenic	Seek help from health worker for persistent sores, smelly mouth, white patches, decreased urine output or difficult of swallowing.

Diarrhea

Diarrhea is defined as the passage of loose stool three or more times in 24 hours. It can be classified as: **Acute** when the diarrhea lasts for two week or less; **Persistent** when the diarrhea last for two to four weeks or as **Chronic** diarrhea lasting more than four weeks

Table: 23.10: Common cause of diarrhea and its management

Common Causes	**Management**
• Osmotic • (lactose intolerance and laxative) • Motility(drugs) • Secretory (Cholera)	I. **Treating Reversible Causes**. II. **Prevent dehydration** Correct dehydration by ORS, drink extra fluids and intravenous fluids may be appropriate in a patient with severe dehydration secondary to severe acute or Chronic diarrhea. Decisions should be made on a case-by-case basis. III. **Symptomatic treatment** Symptom treatment is recommended: • If the diarrhea becomes chronic and is not helped by specific treatment • Contraindications for anti-diarrheal agent • It should not be used if there is fever or blood in the stool • Should be avoided in children under the age of one year old. For symptom control, start with anti-motility agents such as: • Loperamide 4mg once, then 2mg per loose stool to maximum 16 mg/day, • Oral morphine 2.5–5 mg every4 hours *(alternative)* IV. **Home care** Advise the patient and caregivers to: Boil drinking water and store it in a clean container with a cover. Wash hands with water and soap before eating food and after visiting a toilet. Give the sick person drinks frequently in small amounts, such as porridge, water (with food), other soups, or oral rehydration solution (ORS) but avoid soft drinks. Avoid fatty foods, concentrated fruit juices, alcohol, and coffee.

	Use high-fiber foods such as beans, rice, maize meal, green bananas, whole grain bread and potatoes. o Eat bananas and tomatoes (for their potassium) o Carrot soup helps to replace vitamins and minerals. Carrot soup contains pectin. It soothes the bowels and stimulates the appetite. o Eat 5-6 small meals rather than 3 large ones. o Protect the peri-anal skin from excoriation by using petroleum jelly or aluminum hydroxide and keep it clean and dry. **Seek help from health worker if:** • Vomiting with fever • Blood in the stool • Diarrhea continues more than 5 days • If patient becomes even weaker • If broken skin around the rectal area

Constipation

Constipation usually includes a frequency of fewer than three bowel movements a week, but can also include subjective symptoms such as excessive straining, a sensation of lower abdominal fullness, and hard stools.

Table: 23.11: Common cause of constipation and its management

Common Causes	**Management**
• Low fluid intake • Irregular bowel habit • Low residue and fiber diet • Limited physical activity • In hospital, unfamiliar toilet arrangements (such as the use of bedpans and lack of privacy can also lead to constipation) • Drugs such as opiates • Medical conditions (diabetes, hypothyroidism, hypokalaemia, rectocele, cerebrovascular accidents, and Parkinson's disease)	**I. Treat specific causes** **II.Symptom management** Constipation in persons with chronic diseases is often related to Medications and, thus, preventive measures are the most successful. If non-pharmacological measures are not effective, laxatives are appropriate although hard stool requires a softener and soft stool requires a peristaltic stimulant, a combination of both is better with Fewer side effects. A variety of different laxatives can be used: • Stimulant laxatives, which cause intestinal motility to

• Mechanical obstruction • (ColonCa, Kaposi)	increase. • (Commonly stimulant laxatives include, Bisacodyl, 5-10 mg po.) • Emollient laxatives, such as mineral oil and liquid paraffin, which are given orally or by enema that act by penetrating and softening the stool. • Hyperosmolar agents like lactulose 15-30 ml PO or PR twice daily. *(If available also use polyethylene glycol, non-absorbable sugars, or sorbitol)* • Suppositories or enemas can be given; especially when oral laxatives alone are insufficient. **III. Home care** • Encourage high fluid intake and regular bowel habit • Treat mild constipation by increasing the patient's dietary fiber intake to a minimum of 20–35 grams daily (fruits, vegetables, porridge, and locally available high-fiber food) • Take a tablespoon of vegetable oil before breakfast. • Encourage patients to be mobile and provide them with Privacy when using toilet facilities. • Take preventive measures when constipating drugs like opioids are prescribed. Offer drinks often. • If impacted, gently put petroleum jelly or soapy solution In to the rectum. If the patient cannot do it, the caregiver can help— always use gloves. • Seek help from a trained worker if pain or no stool is passed in 5 days.

23.5. Urinary Symptoms

Urinary Incontinence

- It is the involuntary loss of urine, which may cause hygienic, physical and/or social problems. It is essential to determine the cause and identify those patients where the incontinence is due to an inability to reach the toilet in time.

Table: 23.12. common cause of urinary symptoms and the management

Common Causes	Management

Urge incontinence: sudden, strong desire to void, frequency, loss of moderate to large volume of urine, nocturia and/or enuresis usually present due to urethral inflammation or irritation. **Stress incontinence**: leakage with physical activity (coughing, laughing, lifting etc.), small volume of urine, and intermittent dribbling due to weak pelvic floor. **Overflow incontinence**: strain to void, sense of incomplete emptying, lower abdominal pain/fullness due to urethral obstruction, or bladder under activity (neuropathy). **Functional incontinence**: inability to reach the toilet in time due to physical, psychological or environmental impediment. **Total urethral incontinence**: local incompetence of the urethral sphincter tumor invasion, fistula or surgery. Central loss of sphincter control, confusion/dementia	**Correct reversible factors** • Avoid excessive volume of liquid late in the day, caffeinated drinks, alcohol • Treat medical conditions (delirium, infection, • restricted mobility, impaction, diabetes) • Proper use of medication (carbamazepine, diuretics, • lithium, opioids (overflow) • Correct physical/environmental barriers • Odor: identify cause and treat. **Consider disease-specific palliative therapy** • Total urethral incontinence: Regular toileting, • a female urethral catheter or a male sheath catheter may be tried. • Urge incontinence: Timed voiding and anti-cholinergic drugs may be helpful. • Stress incontinence: Support prosthesis such as a ring-pessary or urethral inserts may be useful. • Overflow incontinence: Catheterization (Permanent or intermittent),urethral stenting or surgical interventions should be explored. Referral to an urologist is essential in this instance. Functional incontinence: surgery, psychological support **Institute non-pharmacological interventions** • Patient education (fluid restriction in the afternoon and evening, early intake hours before bed time) • Lifestyle modification (e.g. decreased caffeine) Behavioural modification: timed voiding • calcInvasive devices (e.g. catheter, stent French Gauge(FG) 16 is the smallest useful size for adults, catheters are most appropriate and should be changed every 3-6 weeks. • External collection of urine (diapers) • meticulous skin protection with barrier creams such as • zinc and castor oil is mandatory. • Intermittent catheterisation together with an • absorbent pad. **Pharmacologic Therapy** Infection: Microbial Culture and Sensitivity and appropriate antibiotic Barrier preparation to protect the skin

	(Zinc and castor oil/Vaseline) Relieve constipation (see constipation section) Review medication for medicines which may exacerbate the problem.Referral to appropriate service/ more experienced clinician: Urologist or Gynaecologist

Hematuria

- **Blood in the urine (Hematuria)** is a frequent presentation in urological disease.
- Ranging from microscopic hematuria, detected by urinalysis, to clearly visible frank hematuria or passage of clots.

Causes	**Management**
• Tumor o Renal, ureteric, bladder, prostate. o UTI o Calculi o Trauma • Drug-induced o Warfarin and NSAIDS causing bleeding tendency o Cyclophosphamide causing hemorrhagic cystitis, glomerulonephritis	• Correct **reversible factors** o Treat underlying cause: Once the cause of hematuria is established to be self-limited o Reassure patient and family o Complete evacuation of clots (irrigation with water or saline through large bore urethral Catheter o Cystourethroscopy for diagnosis and cautery if patient fit enough. o We can prevent Cyclophosphamide associated hematuria by sticking to rehydration protocols. o If bleeding is due to drugs –discontinue the causative drug and refer o More severe haemorrhage: consider referral o Blood transfusion based on need • **Clot retention:** o Evacuate clots using large bore (22Fr) Foley catheter and irrigate with saline 0.9% continuously until urine clears. o Cystoscopic bladder irrigations may be needed, referral may be required. o Percutaneous insertion of a suprapubic catheter is contraindicated • **Consider disease-specific palliative therapy** o Palliative radiotherapy o Involve the urologist early in managing this problem

Bladder pain

- **Bladder pain**: more constant pain, from a dull ache in UTI to acute disabling pain and may be a sign of obstruction and retention or excruciating, transient pain due to bladder spasm or urinary retention

Causes	Management
• Cystitis • Urinary obstruction (calculi, clot, BPH, PUV) • Urinary retention Bladder spasm (irritation hyper excitability of trigon) • Local cancer • Bladder • fibrosis(radiation, TB), • Indwelling catheter • Anxiety	• Treat underlying cause • Reassess the indwelling catheter where applicable as mechanical irritation can be caused by the catheter balloon - change catheter/ reduce volume of the balloon • Catheter slugging with partial retention – bladder washouts or continuous • Bladder irrigation as described in the UTI section • Treat cystitis • Relieve obstruction/instability – especially related to catheter • Surgical intervention- tumor, Posterior Urethral Valve, Benign Prostatic Hyperplasia, stone. • Remove foreign body (catheter or stones) if obstructing bladder outlet • Treat Inflammatory causes • Institute non-pharmacological palliative interventions • Regular toileting • Appropriate fluid intake • Avoiding caffeine and alcohol • Prescribe appropriate first-line treatment • Treat bladder pain: WHO pain ladder • Add treatment for bladder spasms with: o Hyoscine Butylbromide 10-20 mg po, IV, IM or S/C tid. Maximum of 100mg per day – avoid if there is urinary retention o Consider adjuvant/second-line treatment o Intravesical morphine and bupivacaine tid (morphine 10-20mg and 0.5% bupivacaine 10ml diluted in 0.9% saline to 20 ml), instil through an indwelling catheter and clamp for 30 min. o Spinal analgesia, e.g. epidural morphine and 0,5% bupivacaine can be used if setup allows

23.6. Skin Care

Itching

- **Itching (pruritus):** is an irritation in the skin that elicits an urge to scratch.
- It is a problem that can be localized or generalized.
- Sometimes, it may be worse at night. A generalized itch and those without visible lesions are often more difficult to treat than a localized itch and those with visible skin lesions.
- Generalized itching with invisible lesion are usually due to systemic problems or allergies.

Causes	Management
• Infection :(Tinea Scabies, Sexually Transmitted Disease,HIV) • Contact with any skin irritants • Drugs(chemotherapy) • Neuropathies(DM, Herpes, MS) • Metabolic problems (renal failure, liver failure, hyperthyroid) • Dry skin • Sunburn • Insects • Psychological(Anxiety, Stress)	• General management options for itching o Avoid heat and hot water o Moisturise and hydrate dry skin o Apply calamine lotion o Menthol 1% in aqueous cream. o Zinc o To moisturize and hydrate the skin use aqueous cream as a soap substitute and bland bath oils o Apply an emollient (liquid paraffin, coconut oil) immediately after washing • Outpatient medication o Assess for infectious cause – if present, treat. o Consider and treat other underlying cause o If there are multiple skin infections, use a chlorhexidine (0.05%) rinse after bathing. o Consider that this may be the side-effects of medication. o Local steroid creams may be useful if inflammation is present in the absence of any infection- bacterial, fungal, or viral. • Anti-histamines o Cetrizine-10mg po once daily, Loratadine 10mg po daily **(**in children less than 30KG use 5mg po daily) o Chlorphenramine 4 mg twice daily, up to 4 mg every 4–6 hours (maximum 24 mg daily), or another antihistamine, may be useful for severe itching. o Diphenhydramine If the itching still persists, a short-acting antihistamine and a long-acting antihistamine (from different groups) could be combined for better symptom relief. • Home care for itching o If the affected person has dry skin, moisturize with aqueous cream or petroleum jelly mixed with water. Avoid frequent bathing o Use 1 spoon of oil (bath or vegetable) in the bath water when washing. o Apply diluted Chlorhexidine (0.05%) after a bath, medicated soap

Pressure Sores

- **Pressure sores (decubitus ulcers)** are ulcers of the skin which can extend into the subcutaneous tissue caused by ischemia secondary to extrinsic pressure and shearing forces.

- **The most vulnerable areas** are elbows, shoulder blades, spine, buttocks and heels. If the patient is more comfortable lying on his/her side, then special attention should be given to ears, shoulders, hips and knees.

Causes	Management
• **Risk factors** o General debility o Neurological deficit o Reduced mobility o Coma o Incontinence o Cachexia o Dehydration o Anemia o Infection and poor hygiene o Others	• **Prevention of pressure sores** o Inspect the skin every time the patient is moved o The skin should be washed and dried regularly, including bed bath for bed ridden patient o Maintain suppleness of skin by regular massage with skin lotion o Avoid trauma – no restraints, lift patients do not drag them to move them in the bed o Regular positional change, family or hospice/hospital carers should assist in changing the patient's position every 2-4 hrs. depending on patient's risk factors o Keep the bed linen dry and free from creases o Keep the patient well-nourished and well hydrated o Using cushions to support joints helps the patient relax and prevent ligaments from being overstretched. o Special mattress (air or water mattress)to distribute body weight more evenly

Wound Care

- **General wound care:** in palliative care we mainly deal with chronic wounds.
- Chronic wounds are characterized by ischemia, lengthened inflammatory processes, increased protease concentration and reduced level of growth factor activity.

Causes	Management
• Classification o Mechanical (surgical and trauma) o Burns and chemical (thermal, radiological) o Chronic ulcerative wounds (pressure sores, leg ulcers, radiotherapy or malignancies) Post-operative o Factors that delay wound healing o Nutrition deficiency Anemia o Medications that slow wound healing-e.g. steroids, NSAIDs,	• Treat reversible factors o Wound specific treatment - Remove Necrotic Tissue - surgical – tweezers, scissors or a scalpel debridement as appropriate - Gauze soaked in adrenaline 1:1000 - Gentle removal of dressing with normal saline spray or irrigation using a syringe containing warm Normal Saline 0.9% to prevent trauma at dressing changes. • Infection o Irrigate wound with warm NS 0.9% or under running water. o Use antibiotics if there is spreading inflammation, not just a red rim. o Use appropriate antibiotics like Cloxacillin or

Chemotherapy, Immunosuppressants ○ Other: Beta blockers, anticoagulants and phenytoin may delay healing. ○ Radiotherapy depletes dermal fibroblasts ○ locally and total body irradiation ○ Depresses bone marrow which causes minimizing of wound macrophages. ○ Drug and Alcohol abuse cause vascular injury and reduce the immune response ○ Ischemia and reduced blood supply (e.g. pressure) ○ Necrotizing malignancy • Location ○ Pressure sores commonly on heel, sacrum and Buttocks ○ Peripheral areas with poor local circulation ○ Position affects vascularity e.g. wounds over joint areas tend to heal slower. ○ Wound circumference and depth – ruler ○ Wound margins – redness could indicate infection, red/grey could be the result of undermining, white typifies maceration (excess moisture) • Exudate ○ Amount: A sudden increase may indicate infection but the presence of exudates is a necessary part of the healing process and varies during the different phases.	Trimethoprim Sulfamethoxazole or Erythromycin ○ Choosing a dressing ○ Administer appropriate anti-pain during wound care ○ To maintain moisture ○ To add moisture ○ To absorb moisture ○ Protect wound surface ○ Control bacteria - Silver Sulphadiazine cream, Ichthammolglycerine, activated charcoal ± silver • dressings ○ Control odor - activated charcoal, see separate guidelines below

○ Appearance: serous, haemo serous,sanguineous, purulent ○ Odor: an offensive odor usually indicates the presence of high levels of bacteria. • Wound base ○ Dry and necrotic ○ Moist and sloughy ○ Granulating ○ Epithelializing	

Malodourous wounds

- The wounds seen in palliative care patients are often a result of their advanced illness and/or poor physical or social state.
- Wound malodor can be a significant and distressing symptom for the patient, his/her family and care givers. Patients may experience embarrassment, disgust, guilt and shame, which can lead to social isolation and relationship problems.

Management

- **Correct reversible factors**

Malodor is often caused by anaerobic bacterial infection of the necrotic tissue within a fungating wound. Metronidazole has been shown to be effective.

- **Consider disease-specific treatment**

Treatment of lesions depends on the stage of the wound, size and the patient's general health:
Surgery (debulking) of large fungating tumors

- ○ Chemotherapy
- ○ Palliative radiotherapy
- ○ Hormone therapy for responsive tumors in breast cancer (Tamoxifen)

- **Non-pharmacological treatment of wound malodors**
  - ○ Good hygiene:
    - ▪ Regular wound cleaning
    - ▪ Regular bedding and clothing changes
    - ▪ Adequate disposal of soiled dressings
    - ▪ Adequate ventilation
  - ○ Dressings
    - ▪ Use highly absorbent dressings to contain a high level of exudate to control odor.
    - ▪ Preferably use non-adherent dressings (to make regular changing less painful)
  - ○ Debridement
- **Pharmacological treatment of malodor in wounds**
  - ○ Topical (For mildly infected wounds without cellulitis):

        - Metronidazole solution for cleaning a wound: 2liter saline + 10 (500mg) crushed metronidazole tablets to wash wound or area.
        - Metronidazole gel: KY jelly mixed with crushed metronidazole tablets applied to the wound
        - Metronidazole powder: crushed metronidazole tablets applied directly on to the wound
        - Metronidazole cream: aqueous cream mixed with crushed metronidazole tablets
        - Silver Sulphadiazine (Flamazine)
        - Bacitracin, Fucidin or antifungal ointments where appropriate
        - Charcoal dressings
        - Honey and yoghurt
    - Systemic: antibiotics for significant infections (preferably after obtaining a wound swab if possible)
        - Metronidazole 500mg tid/bid pediatric dosage PO until good effect is obtained. (Side-effects: nausea and alcohol intolerance).
        - Other antibiotics: Amoxicillin, Erythromycin, Trimethoprim Sulfamethoxazole .
        - Mono-therapy with Metronidazole is usually sufficient to address clostridial and bacteriods effects but sometimes staphylococcus and pseudomonas may colonize such wounds and need additional antibiotics such as Cloxacillin and/or Aminoglycosides
- Pain: Treat pain appropriately

23.7. Infection prevention and control

- Palliative care services shall operate in accordance with National Infection Prevention and Control Policy and standard guidelines to minimize the risk of infections in patients, families and care providers in order to promote a safe caring environment.
    - **Core infection prevention and control interventions shall include:**
        - Hand hygiene
        - Safe handling and disposal of patient excreta such as faeces, urine, sweat, and waste products
        - Use of personal protective equipment
        - Isolation precautions
        - Aseptic technique
        - Cleaning and disinfection and
        - Sterilization

23.8. Neuropsychiatric Symptoms and their Treatment

- Patients and their families/care givers shall be informed of common neuropsychiatric issues.
- A good assessment is key to support patients. Common neuropsychiatric symptoms in palliative care include anxiety, depression, insomnia and delirium. Each of these areas will be considered and their treatment discussed below.

Anxiety

- Anxiety is defined as a feeling of apprehension and fear characterized by physical symptoms such as palpitations, sweating, and feelings of stress.
- It is common in the terminally ill for a variety of reasons such as the fear of uncontrolled symptoms, fear of dying or being left alone to die.

Symptoms of Anxiety

- Core features**:** Persistently tense and unable to relax, worry, more than normal mood variation, cannot distract self.
- Symptoms**:** Poor concentration, indecisiveness, insomnia, irritability, sweating, tremor, nausea, panic attacks.
- **Assessment to identify cause and severity:**
  - Is it severe?
  - Is it long-standing?
  - Is it alcohol withdrawal?
  - Is it situational?
  - Is it related to a specific fear?
  - Are the family anxious?

Management:

- Correct reversible factors
- Non-pharmacological intervention: Psychological methods for managing anxiety. Psychotherapeutic Interventions

General treatment for most anxiety problems

- Psycho-education: this is the first and most important kind of treatment for all of mental health problems, helping people realize that their condition is common, treatable, and not something to be ashamed of.

  - These sorts of problems are very common
  - It is not a form of weakness – our bodies are designed to have these sorts of reactions
  - Though most of these problems will not completely vanish, they almost always improve, though it can take time.
- Learning skills to reduce the effects of stress is the most effective relief.
- Help people become aware of when they are anxious and what seems to provoke it.
- This seems obvious, but, anxious behaviors can be a habit and not noticed.

  - Trying to do things too fast or doing too many things at the same time
  - Holding the body in a tense position, clenching fists, clenching jaw.
  - For some people, impulsively eating
  - When the patient is aware of being anxious or of cues to anxiety, try a method of relaxation:
    - Take a few slow, deep breaths
    - Have a motto or something to think about that reminds them to be calm

- Count slowly to 10 and then continue with whatever they were doing

- Try 'active coping'- if there is a feared issue or thing, try to work on dealing with it rather than avoiding (but all the while acknowledging that it's hard).
  - For children (or even adults), reward brave behavior

Pharmacological Intervention

Drugs	Dosage	Description
Benzodiazepine		
Diazepam	2 to 5 mg p.r.n., p.o./day	Diazepam has a long half-life and may therefore accumulate and be sedative. It should be possible to give it once a day, at night, although initially it can be given tds (three times a day).
Lorazepam	1 to 2 mg p.r.n., p.o.bid/day	Lorazepam is short-acting, rapidly anxiolytic and less sedating than diazepam. It may be more addictive on a longer-term basis.
Antidepressants		
Amitriptyline	25 to 50 mg po/day	
Imipramine	25 to 50 mg po/day	

Note: Refer if no response

Delirium/Acute confusion

- Delirium is an altered state of mind characterized by confusion of recent onset and variable severity.
- It is the most common and serious neuropsychiatric complication in the patient with advanced illness.
- It is a collective term for the various causes of acute confusion rather than a specific diagnosis.
- There are 4 key features of delirium that need to be present to make the diagnosis:
  - A changed level of consciousness (Difficulty focusing, sustaining attention, agitated, restless or drowsy. Disorientation).
  - A disturbance of the process of thinking/cognition (short term memory loss, disorganized thinking, speaking and problem solving, hallucinations and delusions).
  - The above changes are of recent onset and may fluctuate over a period of hours.
  - There is definite clinical evidence that the disturbance is caused by the abnormal physiology of an underlying general medical condition.
- There are 3 clinical sub-types of delirium;
  - **Hyperactive delirium**: The patient is restless, irritable, agitated and may become aggressive or inappropriate in their behavior.
  - **Hypoactive delirium**: The patient is inactive, disinterested and incoherent.

- **Mixed delirium**: The patient fluctuates between hypo- and hyperactive delirium. This is the most common sub-type (>50%).

Management

- Correct reversible factors
- Consider disease-specific palliative therapy
  - Where appropriate rehydrate patients (orally or by infusion).
  - Review all medications, stop or reduce the dosage of all non-essential drugs and recheck for previous excessive alcohol or illicit drug use.
  - A trial of steroids for suspected brain metastases.
  - Most infections should be appropriately treated unless the patient has signs of impending death (within 24-48 hrs.).
  - Consider using bisphosphonates for hypercalcemia.

Non-pharmacological Interventions

- Calm reassurance of the delirious patient
- Regular orientation for time and place
- Presence of a family member- Limit visitors
- Identify and maintain care giver consistency were possible
- Familiar personal objects or photos
- Encourage walking, or if bedridden, range of motion exercises
- Appropriate lighting at night
- Soothing music
- Gentle back massage and a glass of warm milk rather than a sleeping tablet
- Noise reduction as far as possible
- Optimize vision and hearing (check hearing aid)

Note: Physical restraints are not necessary. They may aggravate the situation and cause injury. Effective calming and if necessary, sedation is possible by means of appropriate medication at effective dosages.

Pharmacological Treatment

Mild delirium without agitation			
Drug	Dose	Frequency	Comment
Haloperidol	0.5 - 2mg PO	Can be given every hour, for up to 3 doses and then should be reviewed. (Notify Dr if 3 doses not effective. This allows a review of the situation so that the dose can be adjusted if needed)	Usual effective dose is 0.5 -2mg/day Maintenance: Previous day's total used given as a single daily dose & the same p.r.n. dose for break though symptoms NB: Medication is not always needed but as agitation may occur unexpectedly in a new environment consider its use for a short period.

Delirium with mild agitation but no aggression:			
Drug	Dose	Frequency	Comment

Haloperidol	1-2 mg IM	Can be given every hour, for up to 3 doses and then should be reviewed. (Notify Dr if 3 doses not effective.)	Usual effective dose is 6-12mg/day Maximum 20mg/day Maintenance: Previous day's total used given as a single or divided dose plus the same p.r.n. dose for break though symptoms
Delirium with agitation, restlessness and aggression:			
Drug	Dose	Frequency	Comment
Haloperidol	3-5mg, SC, IM or IV	Can be given every hour, for up to 3 doses and then should be reviewed (Notify Dr if 3 doses not effective.)	Up to 20 mg IM per day. Occasionally prolongation of the Q-T interval may occur. An aggressive delirious patient may be dangerous and calming the patient must be an urgent priority for all staff. (See second line drugs)

Alternative first line drug:

Drug	Dose	Frequency	Comment
Chlorpromazine	12.5 – 50 mg PO, IM or	Every 2-4 hours p.r.n. x3 (Notify Dr if 3 doses not effective.)	More sedating than haloperidol. May cause hypotension, measure BP

Note: Delirium presents many complex clinical and ethical problems. If the degree of uncertainty about the diagnosis and management is interfering with proper care, consult the advice of a more experienced specialists (Psychiatrists and psychiatric nurses).

Insomnia

- Insomnia is the inability to fall asleep or to remain asleep for an adequate length of time. Insomnia is present when all three of the following criteria are met:
  - A complaint of difficulty initiating sleep, difficulty maintaining sleep, and waking up too early.
  - The above sleep difficulty occurs despite adequate opportunity and circumstances for sleep.
  - The impaired sleep produces deficits in day time functions

Assessment to identify cause and severity of symptoms

- It is important to get a thorough history of the patient's sleep pattern now and before he/she started to have problems.
- Establish the patient's understands about insomnia and their expectations.
- Identify possible medications that the patient is taking that could be causing the insomnia.
- Identify possible emotional stresses that could be causing the insomnia.

- Identify possible symptoms such as pain that may be causing the insomnia

Management

- Correct reversible factors
  - Identify and treat the primary condition; e.g. pain
  - Eliminate disturbance and noise
- Consider disease-specific palliative therapy
  - Treat insomnia with a holistic approach
  - Consider physical, psychosocial and spiritual causes and management options
- Institute non-pharmacological interventions
  - Avoid alcohol and stimulants in the evening
  - Relaxation techniques
  - Massage
  - Warm milk/herbal tea at night
  - Cognitive-behavioral therapy
  - Progressive muscle relaxation therapy
  - Hypnotherapy
  - Sleep hygiene
  - Substitute radio or relaxed reading
  - Avoid large meal near bed time

Pharmacologic Management

- Benzodiazepines have been successfully used for short term insomnia, although there are no systematic studies on long-term use and rare studies in palliative care**.**
  - Diazepam 5 to 10 mg at night
  - Lorazepam 1- 2 mg at night
  - Bromazepam 1.5 to 3 mg PO at night
  - Amitriptyline can also be used to treat short term insomnia.

Depression

- Depressed mood and sadness may be an appropriate response to approaching death for the terminally ill patient.
- It is often under diagnosed and under-treated. It is important to identify depression as conventional treatment achieves a good response in the majority of patients.
- If untreated, depression can result in worsening of symptoms and in social withdrawal and even suicide.

Clinical Features

- Depressed mood
- Loss of interest or pleasure in almost all activities
- Anorexia

- Marked weight loss or gain
- Insomnia / hypersomnia / early morning awakening
- Loss of self esteem
- Feelings of worthlessness / helplessness / guilt
- Poor concentration / indecision
- Thoughts of death or suicide

Common causes

- Uncontrolled pain or other symptoms
- Loss of independence
- Lack of support from family / friends / healthcare workers
- Fear – of dying or death
- Drugs – steroids, diazepam, efavirenz
- Loss of body image, due to disfigurement

Management

- Relieve pain and other distressing symptoms
- Review / modify drug treatment
- Provide counseling, support, reassurance
- Drugs – Treatment of depression can significantly improve quality of life and is as effective in palliative care as in other situations.
    - ***Tricyclics***
        - Amitriptyline 25-75 mg at night
        - Imipramine 25-75 mg per day, if available, is an alternative that might be less sedating
    - ***Selective serotonin re-uptake inhibitors (SSRIs)***
        - Fluoxetine 20 to 40 mg in the morning (morning with meal)
        - Sertraline 50 to 100 mg per day (morning after meal with meal)

Note: Consult psychiatrist if there is no improvement.

- 'Start low, go slow'
- Antidepressants should not be stopped abruptly if at all possible, but withdrawn gradually.

Suicide Ideation

Clinical Features

- The frequency of suicide in the cancer population is higher than in the general population with the highest risk in the months after diagnosis.
- The risk decreases with survival time and is low in the terminal phase.
- Depression is a factor in at least 50% of all suicides.
- Treatment of depression can diminish desire for death.

- Feelings of hopelessness – loss of purpose in living – may be associated with suicide intent independently of depression.
- Completed suicide is rare in cancer patients.
- Suicidal thoughts are common in terminally ill patients but are usually fleeting & are often associated with feelings of loss of control & anxiety about the future.
- The desire to hasten death in not necessarily synonymous with a request to hasten death.

Assessment of patients who may be at risk of suicide

- Depression is common in the general community and more so in patients with advanced illness. Separating the clinically depressed patient from the person overwhelmed or despairing of their illness and circumstances may not be easy.
- Questions that can help assessment:
  4. 'Are you depressed?'
  5. 'Recently have you often been bothered by feeling down, depressed or hopeless?'
  6. 'During the last month have you often been bothered by little interest or pleasure or doing things?'
  7. 'Have you felt things getting too much for you?'
  8. 'Have you been scared you might harm yourself?'
  9. 'Do you feel you have lost purpose for your life?'
- The above have been shown to be reliable indicators of patients who may need further exploration of their depressive symptoms and possible treatment.
- It is also important to reassure patients that many people in their circumstances feel this way at times but these thoughts are often fleeting and diminish with time.

Management

- Treat underlying depression
- For a patient who has persistent suicidal thoughts in the absence of an underlying depression, intervention and support from mental health services may be advisable.

23.9. Psychosocial and Spiritual Care and Support

Social care and support in Palliative care

- Social care is an important part of the inter-disciplinary team within palliative care, offering holistic service to patients and families.
- Social care is potentially a universal service and supports clients facing long-term situations involving life-limiting illness, dying, death, grief, and bereavement.
- Using the expertise of social workers with populations from varying cultures, ages and socio-economic status social workers help patients & families across the life span in coping with trauma, suicide, and death.
- Social work practice addressing palliative and end of life care include; hospitals, hospices,

home care, nursing homes, child welfare, civil society's organizations, faith based organizations and family service agencies.

- Social work to chronically sick people deals with a range of problems associated with patients and of his/her family.
- These include: stigma and discrimination, isolation, economic crisis, housing, daily living, dependency for physical needs, cultural influences, religion, psychosocial issues and orphan care.

Spiritual care

- Spiritual care involves being a compassionate presence to patients even as they suffer. It recognizes that emotional and spiritual healing can take place even though a physical cure is impossible.
- As a patient approaches end of life they often begin to think about the meaning and purpose of life and feel the need to mend broken relationships by forgiving and being forgiven.
- Areas of life that can generate spiritual peace or spiritual distress are relationship with God/Creator/Higher Being, with self, with others, and with the world around them.
- Spirituality can be defined completely by the individual's culture.
- Spirituality can also be individually defined by personal experiences unrelated to the culture.

Why is spirituality important?

- Physical healing and psychological coping may be complicated if patients are experiencing spiritual distress.
- Appropriately addressing patients' spiritual concerns and needs can contribute to more rapid recovery and better prognosis.
- Spirituality can bring an ill person hope, strength, and emotional support

Role of communication in Palliative care

- The foundation of palliative care is built on good communication with patients, family members and health care professionals.
- People living with a life-threatening illness often have many concerns about their care and feel isolated.
- The palliative care professional needs to spend time to discuss the patient's problems which may include psychosocial and spiritual problems as well as their physical symptoms.
- When discussing a patient's diagnosis or prognosis the palliative care professional needs to be able to communicate effectively with the patient and family.
- This may take some time and requires a step-by-step approach. The palliative care professional needs to assess the patient and family's desire to know about the life-threatening illness and ensure that the patient and family receive and understand the information as they request and is culturally appropriate.

Care of the care givers